Fake News Classification and Generalization Analysis.

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line 1: 2nd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCI: 5th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 3rd Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 4th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

line 1: 6th Given Name Surname
line 2: *dept. name of organization (of Affiliation)*
line 3: *name of organization (of Affiliation)*
line 4: City, Country
line 5: email address or ORCID

***Abstract*— The spread of misinformation on the Internet has been recognized as a significant problem facing society and therefore the need to automatically detect fake news has been recognized as a significant problem in the field of artificial intelligence. This work attempts to evaluate the performance of various machine learning models in detecting fake news and the work will be based on two publicly available data sets: the WELFAKE data set and the Fake.csv and True.csv data sets. Three models will be employed: the transformer model, which will be implemented as a BERT model; the deep learning model, which will be implemented as a long short-term memory (LSTM) model; and the classical model, which will be implemented as a logistic regression model. Several techniques will be applied to make sure the models don't overfit. The models were evaluated and tested based on various performance metrics, such as precision, recall, F1-score (macro and weighted), ROC-AUC, Matthews Correlation Coefficient (MCC), and Cohen's Kappa. Moreover, interpretable AI methods such as LIME and SHAP were also used to analyze the results and identify text features that influence fake news classification. news. Based on experimental results, the performance of transformer models is proven to be better than that of classical and deep learning models due to their ability to establish contextual relationships in the text. This demonstrates the need to use sophisticated natural language processing and interpretable AI approaches for fake news detection.**

Keywords*— *Transformer model, LIME, SHAP.

I. Justification of Dataset Selection

The datasets used in this study are the WELFake dataset and the Fake.csv + True.csv dataset, both widely used in fake news detection research.

A sizable dataset called WELFake includes both real and fraudulent news articles. The dataset is regarded as a huge dataset and was gathered from multiple sources. It includes textual information, including article titles and content. This makes it appropriate for machine learning techniques and natural language processing. The diversified dataset aids in training models on the traits of false information.

The data of the tagged false and real news articles may be found in the data sets false.csv and True.csv. This information was gathered from a number of news websites. In order to classify fake news items, these data sets are typically utilized as benchmark data sets. A balanced data set for the supervised learning model can be produced by combining these two data sets. Reason of use this dataset:

• For the purpose of training the model, both of these datasets have a substantial amount of labeled data in the form of news articles. The model can learn from the real-world data found in each of these datasets.

• Since both of these datasets are frequently utilized in scholarly articles, comparing the outcomes from both datasets may be simpler.

• The benefit of utilizing these two datasets is that their differences allow the model to be evaluated on data from various sources, increasing its robustness.

Therefore, these datasets can be used to evaluate classical, deep, and transformer-based models for detecting fake news.

A. Dataset Bias

Both datasets have certain biases that may impact the models' performance, notwithstanding their value. Source bias is one possible prejudice. While genuine news stories frequently originate from reliable media institutions, many of the false news articles in the dataset originate from particular websites or political sources. Therefore, instead of recognizing false information, the models can learn to recognize writing styles or source kinds.

Topic bias is another type of bias. Fake news pieces tend to focus more on certain subjects, such international events or politics. This may lead the model to link some subjects, even if they are true, to bogus news. Furthermore, because the datasets include articles that were gathered throughout specific time periods, temporal bias can be present. Models trained on these datasets might not perform well when applied to future news data because news language and disinformation tactics evolve over time..

B. Dataset Noise

However, both of these data sets are still vulnerable to several biases. One of them is source bias. In both data sets, most fake news articles come from a specific source, which is either a political source or a misinformation source, while most real news articles come from a different source. This can

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eventually lead to a situation where the model is learning to identify sources rather than identifying misinformation. Another bias is thematic bias. In this case, there are many topics that are present in fake news articles but not in real news articles. This can eventually lead to a situation where the model is not learning to detect misinformation but rather learning to detect topics. Another bias is temporal bias, which may also be present in this data set. The nature of news and misinformation is constantly changing.

C. Dataset Limitations

These datasets have certain limitations even if they are useful for researching the detection of fake news. First of all, it is discovered that these datasets are limited to English news items, which makes it challenging to apply these trained models to other languages as well. Second, these datasets only contain text-based false information, but there are many other forms of false information available today, such as photos and videos. Thirdly, it's unclear if all forms of disinformation, including satire, propaganda, and partial misinformation, are the foundation of these databases. Fourth, because these datasets are typically brief and based on social media, it is unclear if they are indicative of real news stories or not.

II. RESULT

In this result section we discuss about algorithm such as transformer ,machine learning and deep learning result and comparison for both dataset.

TABLE I. :WELFake DASET-MODEL PERFORMANCE

Model	Precision	Recall	F1-Score	ROC-AUC	MCC	Cohen's Kappa
Logistic Regression	0.93	0.93	0.93	0.97	0.85	0.85
LSTM (Deep Learning)	0.9165	0.91	0.91	0.97	0.83	0.83
BERT	0.99	0.99	0.99	0.99	0.97	0.97

A. Result Interpretation for WELFake Dataset

The table 1 shows that F1-score of 0.99 and ROC-AUC of 0.9994 on the WELFake dataset, the BERT model once again demonstrated the best performance, demonstrating its potent ability to identify semantic links in textual data. containing an F1-score of 0.93, Logistic Regression demonstrated competitive performance, demonstrating that traditional machine learning techniques are still useful in datasets containing significant lexical patterns. With an F1-score of 0.9154, the LSTM model performed well but was marginally less accurate than the other models.

TABLE II. FAKE NEWS DATASET (FAKE.csv + TRUE.csv):

Model	Precision	Recall	F1-Score	ROC-AUC	MCC	Cohen's Kappa
Logistic Regression (TF-IDF)	0.98	0.98	0.98	0.99	0.96	0.96
LSTM	0.9748	0.97	0.97	0.99	0.94	0.9483
BERT	1.00	1.00	1.00	1.00	0.99	0.99

B. Fake News Dataset (Fake.csv + True.csv):

In the table II the experimental findings on the Fake News dataset show that all three models are highly effective at differentiating between real and fake news. Logistic Regression demonstrated that TF-IDF features may identify discriminative word patterns in news articles with an F1-score of 0.98. The LSTM model showed that it could extract sequential contextual information from textual input with an F1-score of 0.9742. The BERT transformer model showed the effectiveness of contextual embeddings and attention mechanisms for classifying fake news with an F1-score of 1.00, ROC-AUC of 1.00, and the highest MCC and Kappa values.

C. Final Comparison Table (Two Datasets)

TABLE III. : DATASET COMAPISON RESULT

Model	Dataset	Precision	Recall	F1-Score	ROC-AUC	MCC
Logistic Regression	Fake.csv + True.csv dataset	0.98	0.98	0.98	0.99	0.96
LSTM	Fake.csv + True.csv dataset	0.9748	0.97	0.97	0.99	0.94
BERT	Fake.csv + True.csv dataset	1.00	1.00	1.00	1.00	0.99
Logistic Regression	WELFake Dataset	0.93	0.93	0.93	0.97	0.85
LSTM	WELFake Dataset	0.9165	0.91	0.91	0.97	0.83
BERT	WELFake Dataset	0.99	0.99	0.99	0.99	0.97

In the table III shows that the comparisons of both datasets show that transformer-based models consistently outperform classical machine learning and deep learning models. While logistic regression provides a robust baseline and LSTM captures sequential patterns in text, BERT uses contextual embedding and self-attention mechanisms to achieve the highest performance in the task of fake news detection.

D. Generalization Experiment:

TABLE IV. GERALIZATION EXPERIMENT COMPARISON

Training Dataset	**Testing Dataset**	**Precision (Macro)**	**Recall (Macro)**	**F1-score (Macro)**	**Precision (Weighted)**	**Recall (Weighted)**	**F1-score (Weighted)**	**Accuracy**
Dataset A	Dataset B	0.03	0.03	0.03	0.03	0.03	0.03	0.03

The table IV shows that cross-dataset generalization result is very low, which is actually a valid research observation. When a model trained on one dataset is tested on a completely different dataset, performance can drop dramatically due to domain shift and feature mismatch.

E. Class-wise Performance:

TABLE V. PERFORMANCE ANALYSIS (CLASS WISE)

Class	**Precision**	**Recall**	**F1-score**	**Support**
Fake (0)	0.06	0.06	0.06	10,454
Real (1)	0.01	0.01	0.01	9,546

This table v clearly shows severe performance degradation when the model is trained on one dataset and tested on another, indicating poor cross-dataset generalization.

DISCUSSION

In conclusion, it can be said that the WELFake dataset and the Fake.csv + True.csv dataset are suitable data sources for conducting research on fake news detection using machine learning algorithms, as they provide a large amount of labeled textual data for training and testing machine learning algorithms. However, it is important to note that these data sources contain potential bias and noise, and future research can work to improve this by using different data sources and multiple sources of information.

REFERENCES